

CSA3006 - DATA MINING AND DATA WAREHOUSING

Presented By

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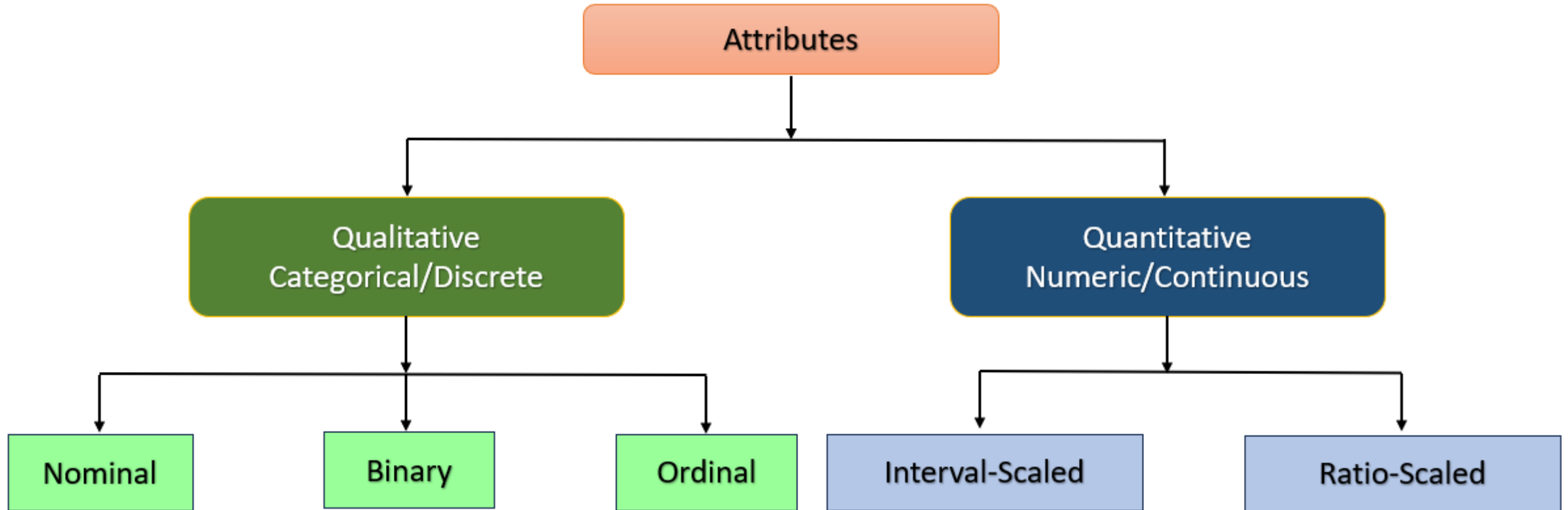
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Module II

- **Data Preprocessing:** Data Objects and attribute types - Basic statistical description of data - Data visualization – Measuring data similarity and dissimilarity - Data cleaning - Integration - Data reduction – Data transformation and data discretization.
- **Association Rule Mining:** Basic concepts - Frequent itemset mining methods - Apriori algorithm, Pattern growth approach for mining frequent itemsets, Mining frequent itemsets using vertical data format, Mining closed and max patterns - Pattern mining in multilevel and multidimensional space.

Data Objects and attribute types



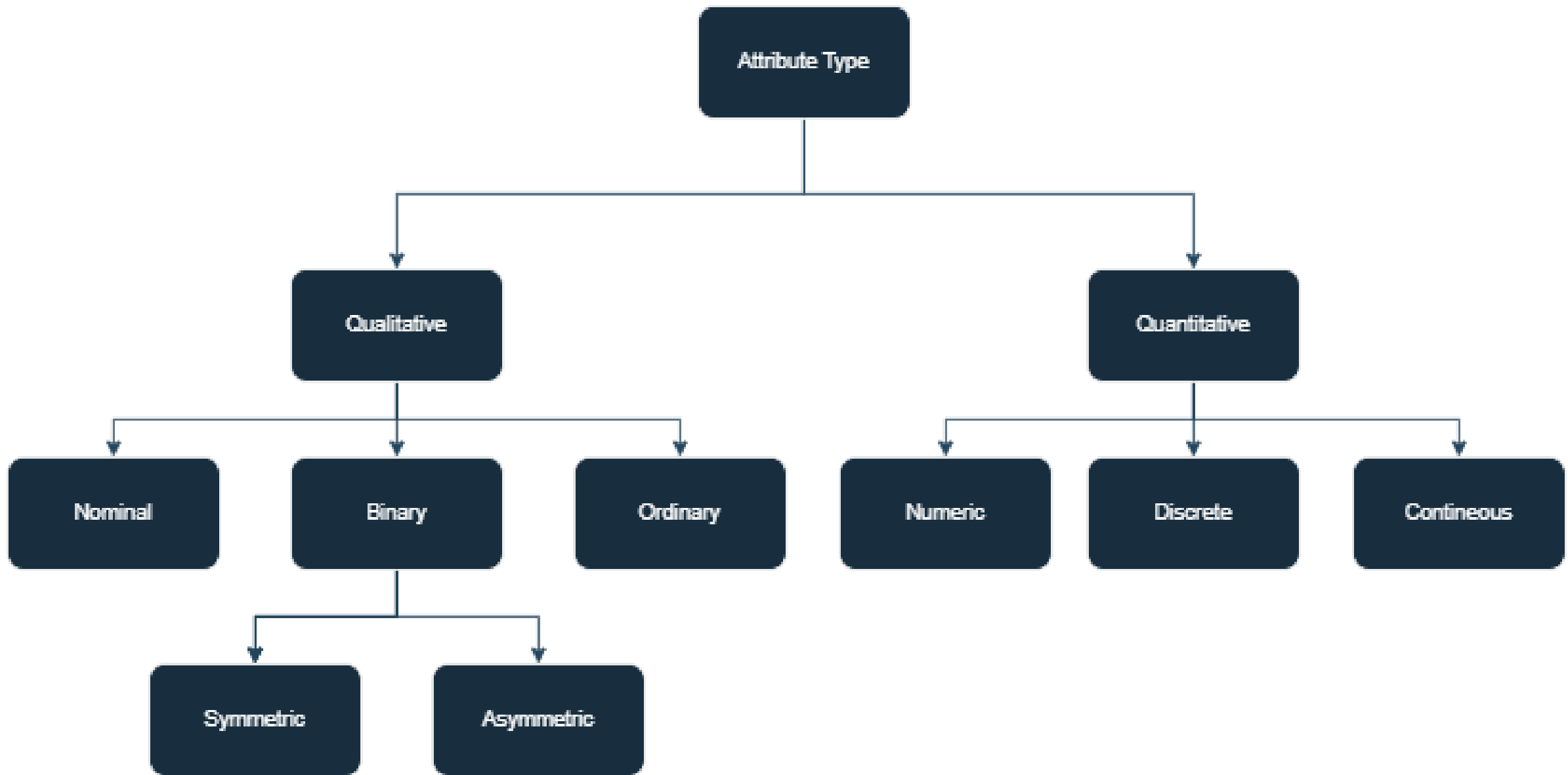
What are Data Attributes?

Data attributes are the characteristics or properties that describe a data object.

- They provide **useful information** about objects in a dataset.
- Attributes are used to **analyze, classify, and process data**.
- Understanding attributes is important in **statistics, machine learning, and data analysis**.
- In **predictive models**, attributes act as **predictors** that influence outcomes.
- In **descriptive models**, attributes are examined to find **patterns or relationships**.
- A group of attributes describing one object is called an **attribute vector** or **feature vector**.
- Examples of data attributes:
 - **Numerical**: age, height, salary
 - **Categorical**: color, type, gender
 - **Textual**: name, description
 - Any **measurable or qualitative** property of data objects.

Data Objects

- **Data objects** are the essential components of a **database**.
- A **data object represents an entity** (such as a person, product, or transaction).
- A data object is a **collection (group) of attributes** that describe an entity.
- Examples of data objects in sales data:
 - **Customer**
 - **Sales**
 - **Purchase**
- When data objects are stored or listed in a **database table**, they are called **data tuples** (or records).



- In **data mining**, understanding **data categories** and **types of attributes** is essential because the choice of mining techniques depends on the **nature of data**.

1. Data Categories

- **1. Structured Data**
- Organized in a **fixed format** (tables, rows, columns).
- Easy to store and query using databases.
- Example: Student database, sales records.

2. Semi-Structured Data

- Does not follow a strict table structure but has **tags or markers**.
- Flexible and self-describing.
- Example: XML, JSON, web logs.

3. Unstructured Data

- No predefined format.
- Requires preprocessing for mining.
- Example: Text documents, images, videos, audio files.

2. Types of Attributes

- An **attribute** is a data **field or feature** that describes an object.

1. Nominal Attributes

- Categorical values with **no inherent order**.
- Only equality comparison is possible.
- Example: Gender (Male/Female), Blood group.

Attribute	Values
Colours	Black, Brown, White
Categorical Data	Lecturer, Professor, Assistant Professor

2. Binary Attributes

- Special case of nominal attributes.
- Have **only two possible values**.
- Types:
 - Symmetric (0 and 1 equally important)
 - Asymmetric (presence more important than absence)
- Example: Yes/No, True/False.

Attribute	Values
Gender	Male , Female

Attribute	Values
Cancer detected	Yes, No
result	Pass , Fail

3. Ordinal Attributes

- Categorical values with a **meaningful order**.
- Differences between values are not measurable.
- Example: Rank (1st, 2nd, 3rd), Grade (A, B, C).

Attribute	Value
Grade	A,B,C,D,E,F
Basic pay scale	16,17,18

4. Numeric Attributes

- Represent **quantitative values**.
- Can be measured and compared mathematically.

a) Interval-Scaled Attributes

- Equal intervals between values.
- No true zero point.
- Example: Temperature in Celsius, Calendar dates.

b) Ratio-Scaled Attributes

- Equal intervals with a **true zero**.
- Ratios are meaningful.
- Example: Height, Weight, Age, Salary.

3. Discrete vs Continuous Attributes

1. Discrete Attributes

- Take **finite or countable values**.
- Example: Number of students, number of visits.

Attribute	Value
Profession	Teacher, Business man, Peon
ZIP Code	301701, 110040

2. Continuous Attributes

- Take **any value within a range**.
- Example: Time, temperature, distance.

Attribute	Value
Height	5.4, 6.2 ...etc
weight	50.33etc

Aspect	Nominal Variables	Ordinal Variables	Continuous Variables	Discrete Variables
Order	No inherent order	Has a meaningful order	Not necessarily ordered	Can be ordered, but takes specific values
Comparison	Can only check equality (same/different)	Can compare ranks (higher/lower)	Intervals are meaningful, arithmetic applies	Intervals are meaningful, but values are finite
Examples	Gender, Eye Color, Country	Satisfaction Levels, Education Levels, Ratings	Height, Weight, Temperature	Number of Children, Goals Scored
Measures of Central Tendency	Mode	Mode, Median	Mode, Median, Mean	Mode, Median, Mean

Basic statistical description of data

- **Statistics** is the **foundation of data science**.
- **Descriptive statistics** are simple methods used to **understand and summarize data**.
- They help to **describe the basic features** of a dataset.
- Common measures include:
 - **Average (mean)**
 - **Highest value (maximum)**
 - **Lowest value (minimum)**
 - **Spread of data (range or variation)**
- Descriptive statistics are the **first step in analyzing and interpreting data**.

Descriptive Statistics

```
graph TD; A[Descriptive Statistics] --> B[Measure of Central Tendency]; A --> C[Measure of Variability]; A --> D[Measures of Frequency Distribution]; B --> B1[Mean]; B --> B2[Mode]; B --> B3[Median]; C --> C1[Range]; C --> C2[Variance]; C --> C3[Standard Diviation]; C --> C4[Interquartile Range (IQR)]; D --> D1[Histogram]; D --> D2[Boxplot]; D --> D3[Pie chart];
```

Measure of Central Tendency

Mean

Mode

Median

Measure of Variability

Range

Variance

Standard Diviation

Interquartile Range (IQR)

Measures of Frequency Distribution

Histogram

Boxplot

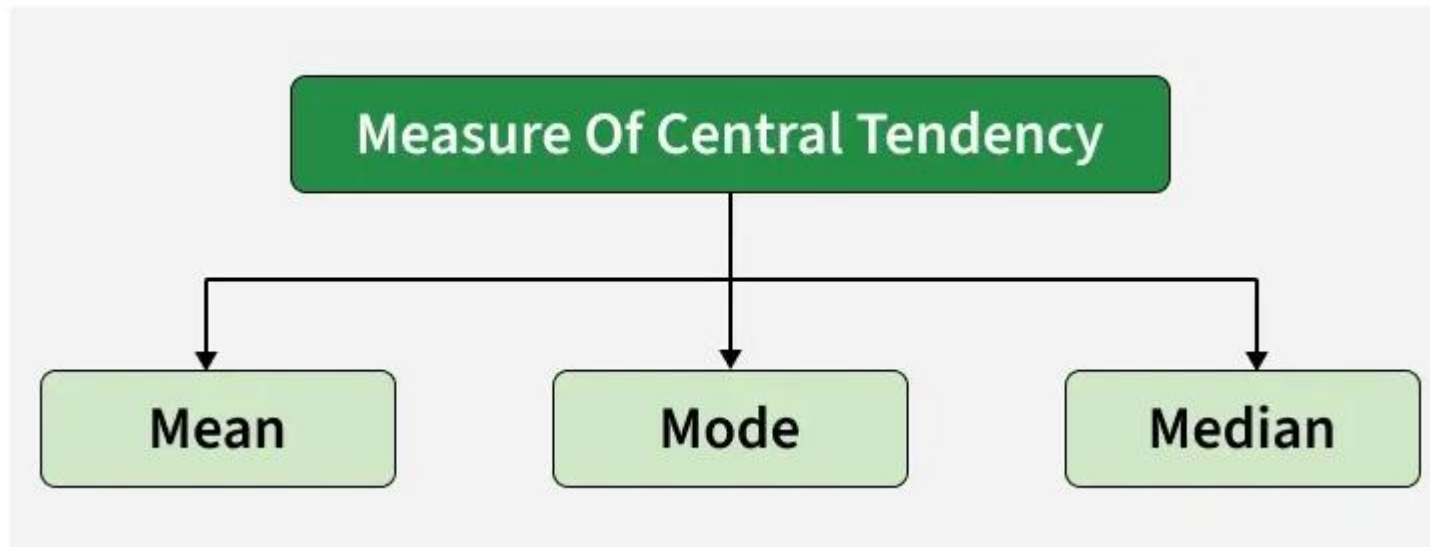
Pie chart

Types of Descriptive Statistics

- There are three categories for standard classification of descriptive statistics methods, each serving different purposes in summarizing and describing data.
- Where the data centers ([Measures of Central Tendency](#))
- How spread out the data is ([Measure of Variability](#))
- How the data is distributed ([Measures of Frequency Distribution](#))

1. Measures of Central Tendency

- Statistical values that describe the central position within a dataset. There are three main measures of central tendency:



Mean: The mean is the sum of all values divided by the number of values. where,

- x = Observations
- n = number of terms

$$\bar{x} = \frac{\sum x}{n}$$

```
import numpy as np

# Sample Data
arr = [5, 6, 11]

# Mean
mean = np.mean(arr)

print("Mean = ", mean)
```

Output

```
Mean = 7.333333333333333
```

Mode: The mode is the **most frequently occurring value** in the dataset.

```
import scipy.stats as stats
```

```
# sample Data
```

```
arr = [1, 2, 2, 3]
```

```
# Mode
```

```
mode = stats.mode(arr)
```

```
print("Mode = ", mode)
```

Output:

```
Mode = ModeResult(mode=array([2]), count=array([2]))
```

Median: The median is the **middle value** when data is arranged in ascending or descending order.

```
# sample Data
arr = [1, 2, 3, 4]

# Median
median = np.median(arr)

print("Median = ", median)
```

Output

```
Median = 2.5
```

2. Measure of Dispersion

These describe how spread out the data is.

a) Range

Difference between the maximum and minimum values.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

b) Variance

Measures how far data points are from the mean.

c) Standard Deviation

Square root of variance; indicates the average spread of values from the mean.

Range = Largest data value - smallest data value

```
import numpy as np

# Sample Data
arr = [1, 2, 3, 4, 5]

# Finding Max
Maximum = max(arr)
# Finding Min
Minimum = min(arr)

# Difference Of Max and Min
Range = Maximum-Minimum
print("Maximum = {}, Minimum = {} and Range = {}".format(
    Maximum, Minimum, Range))
```

Output

```
Maximum = 5, Minimum = 1 and Range = 4
```

3. Measures of Position

- These describe the **relative standing** of a value in a dataset.
- Quartiles (Q1, Q2, Q3)
- Percentiles
- Example:
Q2 is the **median**.

4. Measures of Shape

- These describe the **distribution pattern** of data.
- **Skewness** – indicates asymmetry of data
 - Positive skew: tail on right
 - Negative skew: tail on left
- **Kurtosis** – indicates peakedness of data.

Dataset: The following table shows the **marks obtained by 10 students** in a test:

Student	Marks
S1	40
S2	45
S3	50
S4	55
S5	60
S6	60
S7	65
S8	70
S9	75
S10	80

Based on the above dataset, calculate:

1. Mean
2. Median
3. Mode
4. Range
5. Variance
6. Standard Deviation
7. First Quartile (Q1) and Third Quartile (Q3)
8. Identify whether the data is positively or negatively skewed (basic interpretation).

Step 1: Mean

$$\text{Mean} = \frac{\sum X}{N}$$

Sum of marks:

$$40 + 45 + 50 + 55 + 60 + 60 + 65 + 70 + 75 + 80 = 600$$

$$\text{Mean} = \frac{600}{10} = 60$$

Step 2: Median

Number of observations = 10 (even number)

Median = Average of 5th and 6th values

$$\text{Median} = \frac{60 + 60}{2} = 60$$

Step 3: Mode

Most frequently occurring value = **60**

Mode = **60**

Step 3: Mode

Most frequently occurring value = **60**

Mode = **60**

Step 4: Range

Range = Maximum – Minimum

$$80 - 40 = 40$$

Range = **40**

Step 5: Variance

Formula:

$$\text{Variance} = \frac{\sum(X - \bar{X})^2}{N}$$

Mean = 60

Marks (X)	X – Mean	(X – Mean) ²
40	-20	400
45	-15	225
50	-10	100
55	-5	25
60	0	0
60	0	0
65	5	25
70	10	100
75	15	225
80	20	400

Sum of squares = 1500

$$\text{Variance} = \frac{1500}{10} = 150$$

Step 6: Standard Deviation

$$\text{Standard Deviation} = \sqrt{150} \approx 12.25$$

Standard Deviation ≈ 12.25

Step 7: Quartiles

Data is already ordered.

Q1 = Median of first half

First half: 40, 45, 50, 55, 60

Q1 = 50

Q3 = Median of second half

Second half: 60, 65, 70, 75, 80

Q3 = 70

Step 8: Skewness (Basic Interpretation)

Mean = 60

Median = 60

Mode = 60

Since Mean $\approx$ Median $\approx$ Mode

→ Data is **approximately symmetric** (no skewness)

Data Visualization

- **Data visualization** is the graphical representation of data using charts, graphs, or diagrams. It helps to:
- Understand patterns and trends
- Compare values easily
- Identify outliers
- Communicate insights clearly

Common Types of Data Visualization

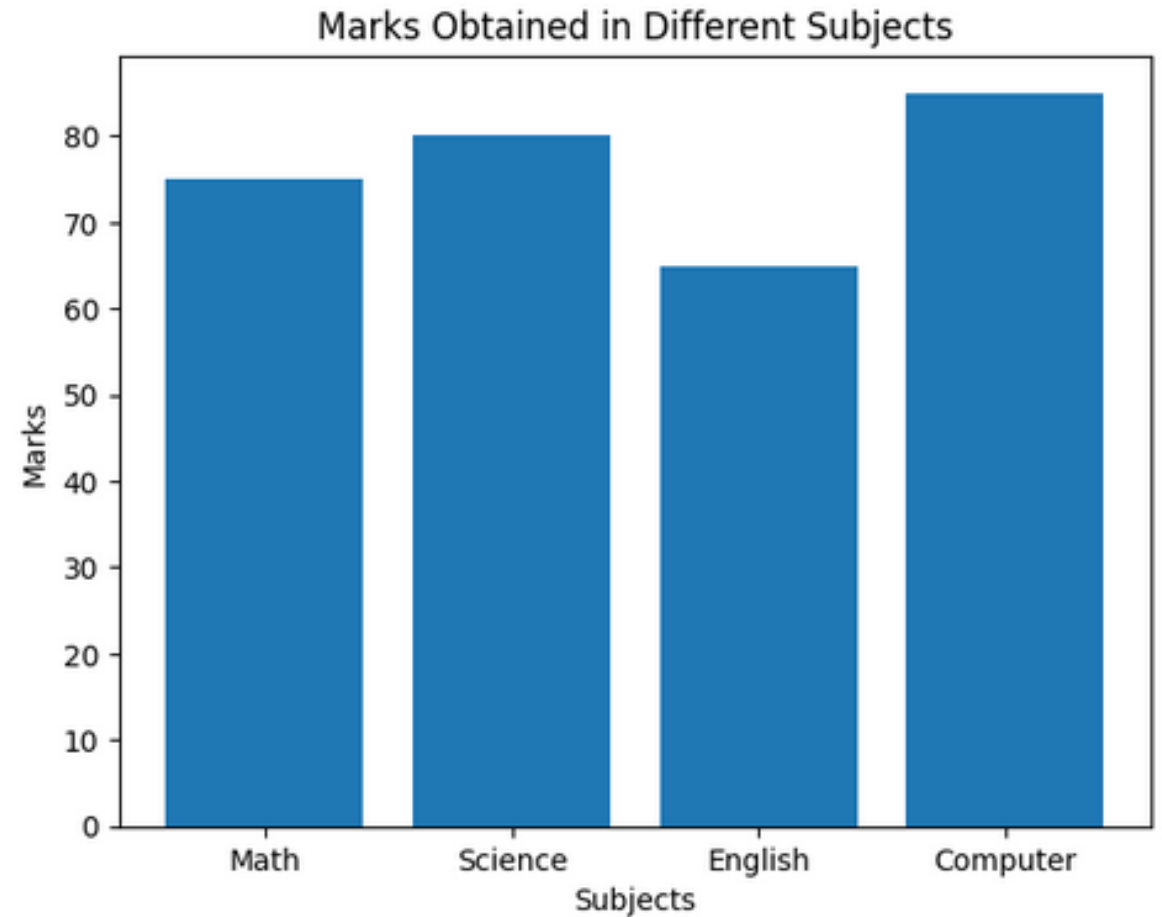
1. Bar Chart

- A **bar chart** is used to compare values across different categories.
- **Comparing quantities** (marks, sales, counts).
- **Example:** Comparing marks of students in different subjects.

Subject	Marks
Math	75
Science	80
English	65
Computer	85

In a bar chart:

- X-axis → Subjects
- Y-axis → Marks
- Height of bars represents marks



2. Pie Chart

- A **pie chart** shows proportions or percentages of a whole.
- Showing percentage distribution.
- **Example:** Time spent on activities in a day.

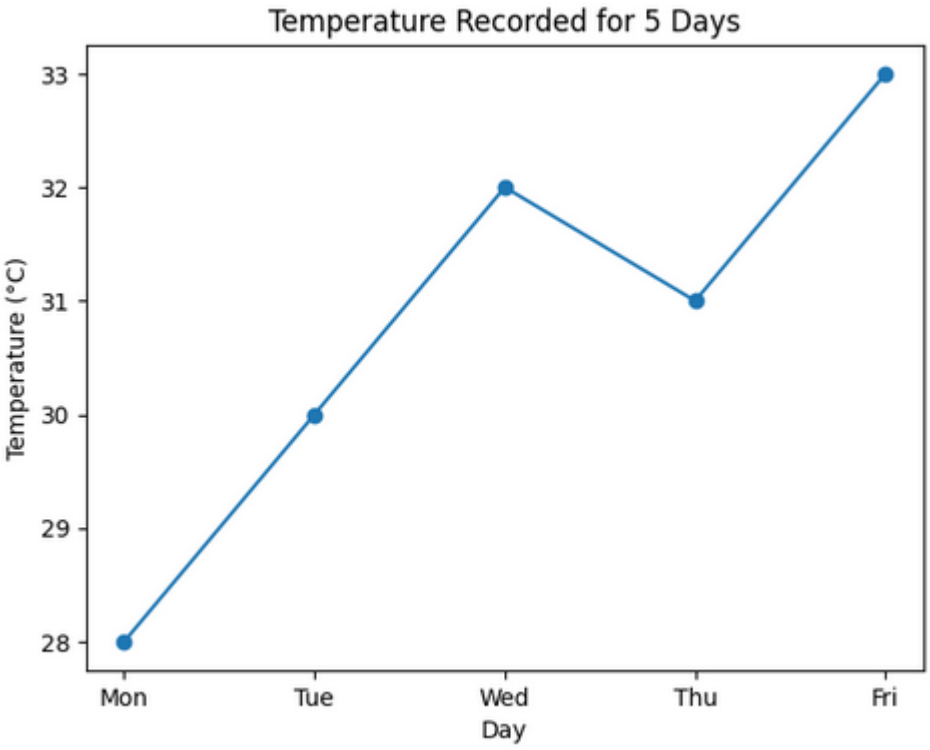
Activity	Hours
Study	6
Sleep	8
Exercise	2
Leisure	4
Others	4



3. Line Chart

- A **line chart** is used to show trends over time.
- **Data** changes continuously (temperature, stock prices, attendance).
- **Example:** Temperature recorded for 5 days.

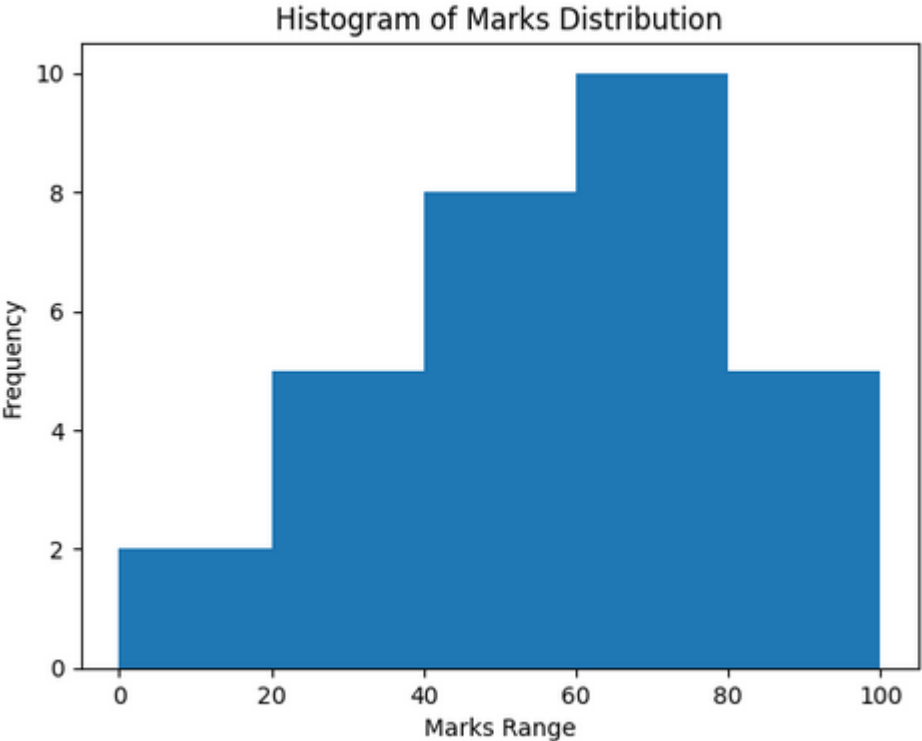
Day	Temperature (°C)
Mon	28
Tue	30
Wed	32
Thu	31
Fri	33



4. Histogram

- A **histogram** represents the frequency distribution of numerical data.
- **Data** is continuous and grouped into intervals.
- **Example:** Marks distribution

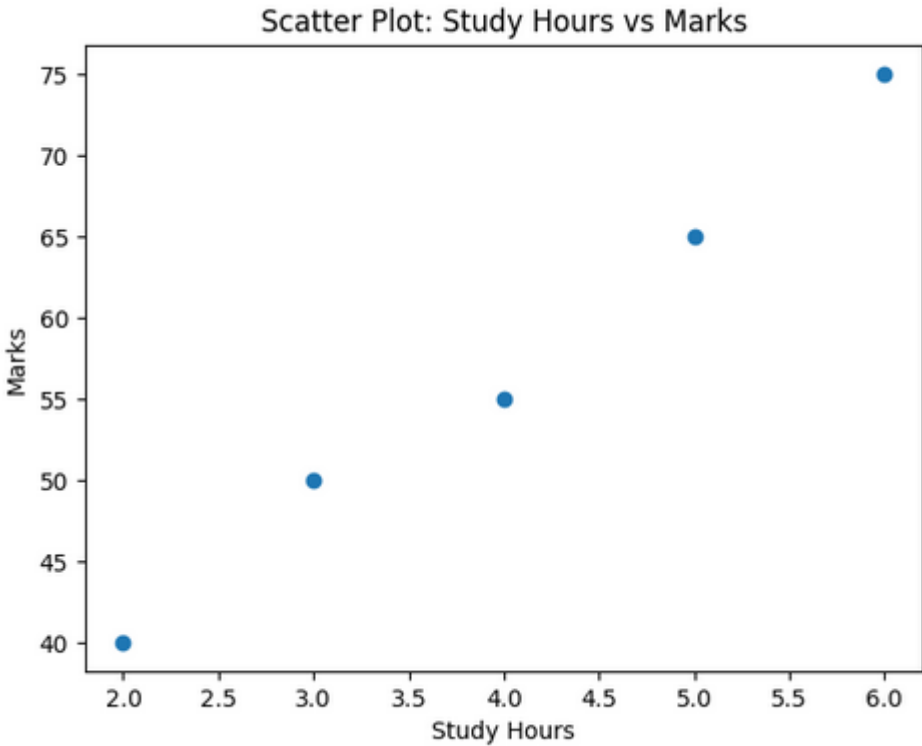
Marks Range	Frequency
0–20	2
20–40	5
40–60	8
60–80	10
80–100	5



5. Scatter Plot

- A **scatter plot** shows the relationship between two variables.
- Studying correlation.
- **Example:** Study hours vs marks

Study Hours	Marks
2	40
3	50
4	55
5	65
6	75



Measuring data similarity and dissimilarity

In data analysis and data mining, **similarity** measures how much two objects are alike, while **dissimilarity** measures how different they are.

- High similarity → Objects are similar
- High dissimilarity → Objects are different
- ***These measures are widely used in:***
 - Clustering
 - Classification
 - Recommendation systems
 - Pattern recognition

1. Distance-Based Measures (Dissimilarity)

- Distance measures how far two data points are from each other.

(a) Euclidean Distance

- Most common distance measure.

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Student	Math	Science
A	50	60
B	55	65

$$d = \sqrt{(50 - 55)^2 + (60 - 65)^2}$$

$$d = \sqrt{25 + 25} = \sqrt{50} = 7.07$$

So, **distance = 7.07** (moderate dissimilarity).

2. Manhattan Distance

- Distance measured along axes (city-block distance).

$$d = |x_1 - x_2| + |y_1 - y_2|$$

Example:

$$d = |50 - 55| + |60 - 65| = 5 + 5 = 10$$

3. Cosine Similarity

- Measures similarity based on the angle between two vectors.

$$\text{Cosine Similarity} = \frac{A \cdot B}{|A||B|}$$

Used in:

- Text mining
- Document similarity
- Recommendation systems

Value range:

- 1 → very similar
- 0 → not similar

Example

Student A marks: (2,3)

Student B marks: (4,6)

$$\text{Cosine} = 1$$

Means perfectly similar (same direction).

$$\begin{aligned} A \cdot B &= (2 \times 4) + (3 \times 6) \\ &= 8 + 18 = 26 \end{aligned}$$

$$\begin{aligned} |A| &= \sqrt{2^2 + 3^2} \\ &= \sqrt{4 + 9} = \sqrt{13} \end{aligned}$$

$$\begin{aligned} |B| &= \sqrt{4^2 + 6^2} \\ &= \sqrt{16 + 36} = \sqrt{52} \\ \sqrt{52} &= \sqrt{4 \times 13} = 2\sqrt{13} \end{aligned}$$

$$\begin{aligned} \text{Cosine Similarity} &= \frac{26}{\sqrt{13} \times 2\sqrt{13}} \\ &= \frac{26}{2 \times 13} \\ &= \frac{26}{26} = 1 \end{aligned}$$

4. Jaccard Similarity (for categorical data)

$$J(A, B) = \frac{A \cap B}{A \cup B}$$

- Example:
- Student A subjects: {Math, Physics, English}
Student B subjects: {Math, English, Biology}
- Intersection = 2
Union = 4
 $J=2/4=0.5$

5. Minkowski Distance

- **Minkowski distance** is a general distance measure used to find **dissimilarity between two data points**.

It is a generalized form of:

- Manhattan Distance
- Euclidean Distance
- By changing the value of **p**, Minkowski distance gives different distance measures.

$$D = \left(\sum |x_i - y_i|^p \right)^{\frac{1}{p}}$$

Where:

- x_i, y_i = values of two data points
- p = order of the distance
- D = distance between two points

- Special Cases of Minkowski Distance

Value of p	Distance Type
$p = 1$	Manhattan Distance
$p = 2$	Euclidean Distance
$p \rightarrow \infty$	Chebyshev Distance

Example

Student	Math	Science
A	10	20
B	13	24

Find Minkowski distance for:

1. $p = 1$
2. $p = 2$

Step 1: Differences

Math difference:

$$|10 - 13| = 3$$

Science difference:

$$|20 - 24| = 4$$

Case 1: $p = 1$ (Manhattan Distance)

$$D = |3| + |4| = 7$$

Distance = 7

Case 2: $p = 2$ (Euclidean Distance)

$$D = \sqrt{3^2 + 4^2}$$

$$D = \sqrt{9 + 16} = \sqrt{25} = 5$$

Distance = 5

Case 3: $p = 3$ (General Minkowski Example)

$$D = (3^3 + 4^3)^{1/3}$$

$$D = (27 + 64)^{1/3}$$

$$D = (91)^{1/3} \approx 4.49$$

Distance $\approx$ **4.49**

6. Hamming Distance

Hamming distance measures the **number of positions at which two strings of equal length differ**.

- *It is commonly used in:*
 - Binary data comparison
 - Error detection and correction
 - DNA sequence comparison
 - Coding theory
- **Idea:**
Count how many positions are different.

Hamming Distance = Number of mismatched positions

Example 1: Binary Data

Two binary strings:

A = **1011101**

B = **1001001**

Compare each position:

Position	A	B	Different ?
1	1	1	No
2	0	0	No
3	1	0	Yes
4	1	1	No
5	1	0	Yes
6	0	0	No
7	1	1	No

Number of differences = **2**

Hamming Distance = 2

5. Similarity vs Dissimilarity Relationship

$$\text{Dissimilarity} = 1 - \text{Similarity}$$

- Example:
Similarity = 0.8
Dissimilarity = 0.2

Measure	Type	Used For
Euclidean	Distance	Numerical data
Manhattan	Distance	Grid-based data
Cosine	Similarity	Text, vectors
Jaccard	Similarity	Categorical data

Data cleaning - Integration - Data reduction – Data transformation and data discretization

1. Data Cleaning

A dataset contains the following student marks (out of 100):
45, 52, 48, NA, 51, 49, 200, 47, NA, 50

Question:

1. Identify:

- Missing values
- Noisy or inconsistent values

2. Replace missing values using the **mean of valid marks**.

3. Remove or correct the noisy value (outlier) and compute the **new mean**.

Step 1: Identify Issues

- Missing values: NA, NA
- Outlier (noisy value): 200 (marks exceed normal range 0–100)

Step 2: Compute Mean (Ignoring NA and Outlier)

- Valid values:
45, 52, 48, 51, 49, 47, 50
- Sum = 342
Number of values = 7
- Mean = $342 / 7 = 48.86 \approx 49$

Step 3: Replace Missing Values

- Replace NA with 49:
- 45, 52, 48, 49, 51, 49, 200, 47, 49, 50

Step 4: Remove Outlier (200)

- Cleaned dataset:
45, 52, 48, 49, 51, 49, 47, 49, 50

Step 5: Compute New Mean

- Sum = 440
Count = 9
- New Mean = $440 / 9 = 48.89$

2. Data Integration

Two departments maintain student records separately.

Table A

Roll No	Name	Marks
101	Ravi	75
102	Meena	82
103	Arjun	68

Table B

Roll No	Attendance (%)
101	88
102	92
103	79

Questions

- 1.Integrate both tables into a single dataset.
- 2.Compute the **average attendance** of students who scored **more than 70 marks**.

Step 1: Merge Tables

Roll No	Name	Marks	Attendance
101	Ravi	75	88
102	Meena	82	92
103	Arjun	68	79

Step 2: Select Students with Marks > 70

- Ravi → Attendance = 88
Meena → Attendance = 92

Step 3: Compute Average Attendance

- Average = $(88 + 92) / 2 = 90\%$

3. Data Reduction

A dataset contains marks of 12 students:

56, 60, 61, 59, 58, 57, 90, 92, 91, 89, 88, 93

Question

1. Apply **binning (bin size = 3)** to reduce the data.
2. Compute:
 - Bin means
 - Bin medians
3. Represent the reduced dataset using **bin means**.

Step 1: Sort Data

56, 57, 58, 59, 60, 61, 88, 89, 90, 91, 92, 93

Step 2: Create Bins (Size = 3)

Bin 1: 56, 57, 58

Bin 2: 59, 60, 61

Bin 3: 88, 89, 90

Bin 4: 91, 92, 93

Step 3: Compute Bin Means

Bin 1 mean = $(56+57+58)/3 = 57$

Bin 2 mean = $(59+60+61)/3 = 60$

Bin 3 mean = $(88+89+90)/3 = 89$

Bin 4 mean = $(91+92+93)/3 = 92$

Step 4: Reduced Dataset (Using Bin Means)

57, 57, 57, 60, 60, 60, 89, 89, 89, 92, 92, 92

4. Data Transformation

The monthly sales of a product (in units) are:

120, 150, 180, 210, 240

Question:

1. Apply **Min-Max normalization** to scale values between **0 and 1** using:

$$v' = \frac{v - \min}{\max - \min}$$

- Range = $240 - 120 = 120$
- For each value:
- $120 \rightarrow (120 - 120) / 120 = 0$
- $150 \rightarrow 30 / 120 = 0.25$
- $180 \rightarrow 60 / 120 = 0.50$
- $210 \rightarrow 90 / 120 = 0.75$
- $240 \rightarrow 120 / 120 = 1$
- Normalized dataset:
0, 0.25, 0.50, 0.75, 1

A. Z-Score Normalization

$$Z = \frac{X - \text{Mean}}{SD}$$

- Mean = 180
SD = 40
- For 150:
 $Z = (150 - 180) / 40 = -30 / 40 = -0.75$ 150 is **0.75 standard deviations below the mean.**
- For 210:
 $Z = (210 - 180) / 40 = 30 / 40 = 0.75$ (210 is **0.75 standard deviations above the mean.**)

Standard Deviation

$$\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{N}}$$

- Where:
- X_i = each value
- μ = mean
- N = number of values

5. Data Discretization

Dataset:

18, 22, 25, 28, 32, 35, 38, 42, 45, 50

Step 1: Find Interval Width

- Min = 18
Max = 50
- Width = $(50 - 18) / 3 = 32 / 3 \approx 10.67$

Intervals:

- $18 - 28.67 \rightarrow$ Young
- $28.68 - 39.34 \rightarrow$ Middle-aged
- $39.35 - 50 \rightarrow$ Senior

Step 2: Assign Values

Young: 18, 22, 25, 28

Middle-aged: 32, 35, 38

Senior: 42, 45, 50

Step 3: Count

Young = 4

Middle-aged = 3

Senior = 3

- **Association Rules:**

- Association rules are data mining rules used **to identify frequent patterns and relationships among items in large datasets.**
- Association rules do **not predict labels** like classification; instead, they **find hidden relationships** in data.
- **Association Rule Mining** finds relationships of the form:
 $X \rightarrow Y$ which means **when X occurs, Y tends to occur as well.**

Example

- If customers buy **Milk**, they often buy **Bread**
Rule: **Milk $\rightarrow$ Bread**
- This rule is evaluated using:
- **Support**
- **Confidence**
- **Lift**

Frequent Itemset Mining Methods

- **Frequent itemset mining** is a key task in **data mining** used to discover sets of items that appear together frequently in a transaction database.

It is widely used in:

- Market basket analysis
- Recommendation systems
- Web usage mining
- Bioinformatics

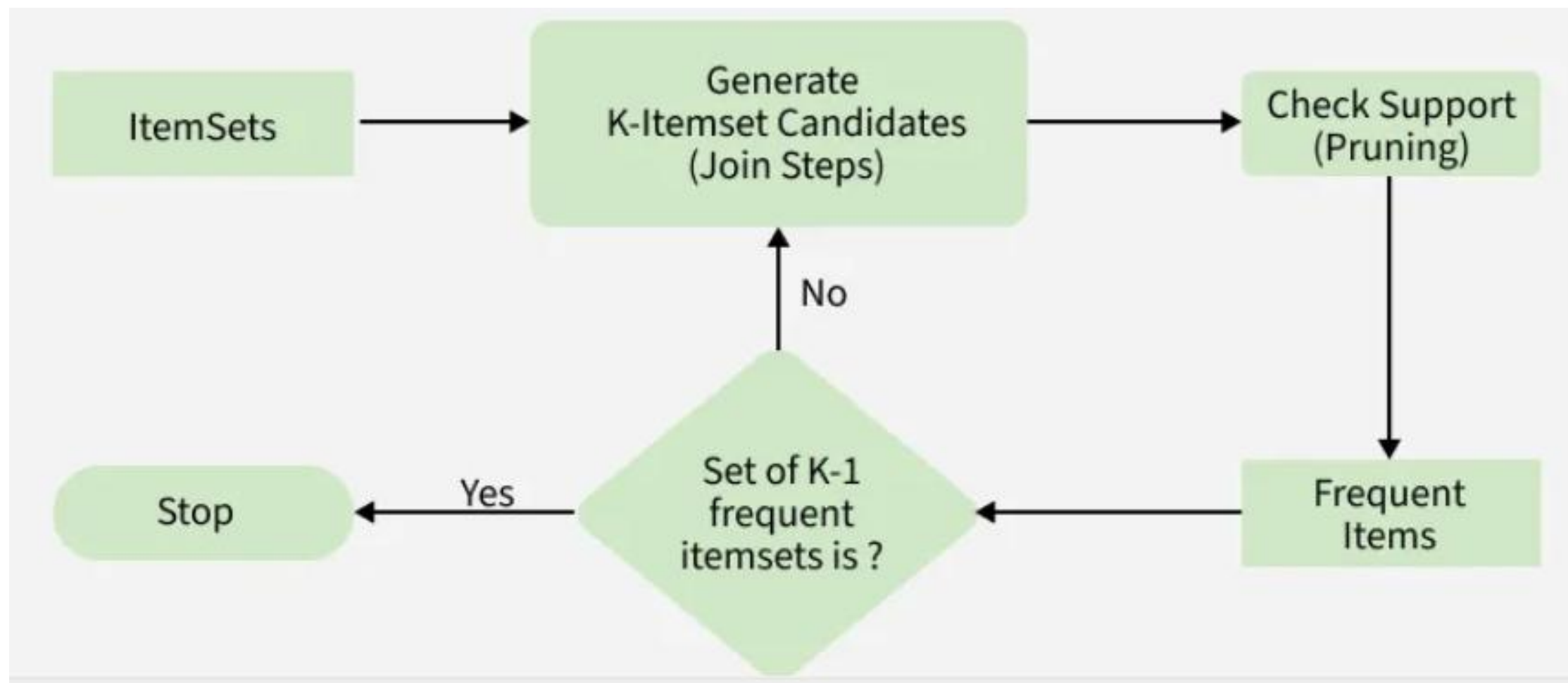
$$Support(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}}$$

- A set of items is called a **frequent itemset** if its **support** is greater than or equal to a specified **minimum support (minsup)**.

Major Frequent Itemset Mining Methods

1. Apriori Algorithm

- **Apriori Algorithm** is a basic method used in **data analysis**.
- It is used to find **groups of items that frequently occur together** in large datasets.
- The algorithm helps to **discover patterns and relationships** between items.
- It is commonly applied in **market basket analysis**.
- Businesses use it to understand **customer purchasing behavior**.
- It works by **identifying frequent itemsets** based on minimum support.
- These frequent itemsets can be used to generate **association rules**.



Key Metrics of Apriori Algorithm

- **Support**
- **Confidence**
- **Lift**
- **Leverage**
- **Conviction**

Apriori Algorithm

Apriori uses the **downward closure property (Apriori property)**:

If an itemset is frequent, all its subsets must also be frequent.

Steps

1. Scan database to find frequent **1-itemsets (L1)**.
2. Generate candidate itemsets **(C2)** from L1.
3. Scan database and count support.
4. Keep only frequent itemsets.
5. Repeat until no more frequent itemsets are found.

Numerical Problem – Apriori Algorithm

A small store records the following transactions:

Transaction ID	Items Purchased
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Milk, Bread, Butter

Minimum Support Count = **2**

Find the **frequent itemsets** using the **Apriori Algorithm**.

Step 1: Find Frequent 1-Itemsets (L1)

Count the support of each item.

Item	Support Count
Milk	4
Bread	4
Butter	4

All items satisfy minimum support (≥ 2).

So,

L1 = {Milk, Bread, Butter}

Step 2: Generate Candidate 2-Itemsets (C2)

Possible pairs:

- Milk, Bread
- Milk, Butter
- Bread, Butter
- Count support:

Itemset	Support Count
Milk, Bread	3
Milk, Butter	3
Bread, Butter	3

All satisfy minimum support.

So,

L2 = {Milk,Bread}, {Milk,Butter}, {Bread,Butter}

Step 3: Generate Candidate 3-Itemsets (C3)

- Possible combination:
- Milk, Bread, Butter
- Count support:

Itemset	Support Count
Milk, Bread, Butter	2

- Support ≥ 2 , so it is frequent.
- So,
L3 = {Milk, Bread, Butter}

- **Step 4: Generate Candidate 4-Itemsets**
- Not possible (only 3 items exist).
- Algorithm stops.

Final Frequent Itemsets

- Frequent 1-itemsets:
- Milk
- Bread
- Butter

Frequent 2-itemsets:

- Milk, Bread
- Milk, Butter
- Bread, Butter

Frequent 3-itemset:

- Milk, Bread, Butter

Largest frequent itemset = Milk, Bread, Butter

2. FP-Growth (Frequent Pattern Growth)

Avoid candidate generation by using a compressed structure called **FP-Tree (Frequent Pattern Tree)**.

- **Steps**

- Scan database and compute item frequencies.
- Build FP-Tree.
- Extract frequent patterns recursively from the tree.

- **Advantages**

- Faster than Apriori.
- Requires fewer database scans.
- Efficient for large datasets.

- **Disadvantages**

- FP-Tree construction can be complex.
- Memory usage may increase for very sparse data.

- Consider the following transaction database:

Transaction ID	Items Purchased
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Milk, Bread, Butter

Minimum Support Count = **2**

Find the **frequent itemsets** using **FP-Growth**.

Step 1: Find Support of Each Item

- Count frequency of each item.

Item	Support Count
Milk	4
Bread	4
Butter	4

All items satisfy minimum support (≥ 2).

Step 2: Sort Items by Frequency

- Since all supports are equal, we keep a fixed order:
- **Milk → Bread → Butter**

Step 3: Reorder Transactions

- Arrange items in each transaction using the order above.

Transaction	Ordered Items
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Milk, Bread, Butter

Step 4: Construct FP-Tree

Insert Transactions into FP-Tree

- Start with a **null root** and insert transactions one by one.

Insert T1 (Milk → Bread → Butter)

```
Null
└─ Milk:1
    └─ Bread:1
        └─ Butter:1
```

Insert T2 (Bread → Butter)

- Bread does not start with Milk, so create a new branch:

```
Null
|— Milk:1
|   └─ Bread:1
|       └─ Butter:1
└─ Bread:1
    └─ Butter:1
```

Insert T3 (Milk → Bread)

- Milk exists → increase count
Bread exists → increase count

```
Null
├── Milk:2
│   └── Bread:2
│       └── Butter:1
└── Bread:1
    └── Butter:1
```

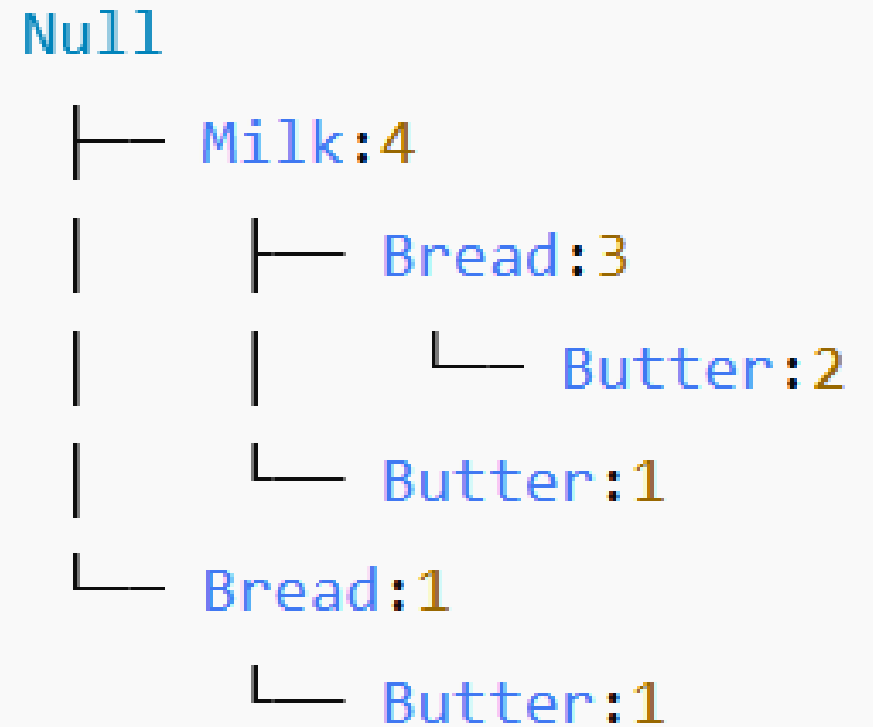
Insert T4 (Milk → Butter)

- Milk exists → increase count
Butter under Milk does not exist directly, so create new node.

```
Null
├─ Milk:3
│   ├─ Bread:2
│   │   └─ Butter:1
│   └─ Butter:1
└─ Bread:1
    └─ Butter:1
```

Insert T5 (Milk → Bread → Butter)

- Milk exists → increase count
Bread exists → increase count
Butter exists → increase count
- **Final FP-Tree**



- Milk appears in T1, T3, T4, T5 $\rightarrow$ count = 4
- Bread under Milk appears in T1, T3, T5 $\rightarrow$ count = 3
- Butter under Milk $\rightarrow$ Bread appears in T1 and T5 $\rightarrow$ count = 2

The FP-Tree:

- Compresses the database.
- Stores common prefixes together.
- Helps mine frequent patterns efficiently.

Mining Frequent Itemsets Using Vertical Data Format (ECLAT)

- Mining frequent itemsets using a **vertical data format** is a method where the database is represented using **TID lists (Transaction ID lists)** instead of listing items in each transaction.
- This approach is commonly used in the **ECLAT (Equivalence Class Transformation)** algorithm.

Vertical Data Format

- In the vertical format:
- Each **item** is stored with a **list of transaction IDs** in which it appears.
- Support is calculated by **intersecting TID lists**.

$$Support(X \cup Y) = |TID(X) \cap TID(Y)|$$

Steps in Vertical Format Mining

- Convert the transaction database into **vertical format**.
- Find frequent **1-itemsets** using minimum support.
- Generate **2-itemsets** by intersecting TID lists.
- Continue generating larger itemsets until no frequent itemsets remain.

Advantages

- Fast support counting.
- Fewer database scans.
- Efficient for dense datasets.

Disadvantages

- High memory usage for large TID lists.

Example

Consider the following transaction database:

Transaction ID	Items
T1	Milk, Bread
T2	Bread, Butter
T3	Milk, Bread, Butter
T4	Milk, Butter
T5	Bread, Butter

Minimum Support Count = **2**

Find the **frequent itemsets using vertical data format.**

Solution

Step 1: Convert to Vertical Format

Item	TID List
Milk	T1, T3, T4
Bread	T1, T2, T3, T5
Butter	T2, T3, T4, T5

Support counts:

- Milk = 3
- Bread = 4
- Butter = 4
- All are frequent.

Step 2: Generate Frequent 2-Itemsets

Milk $\cap$ Bread

- TID(Milk) = T1, T3, T4
TID(Bread) = T1, T2, T3, T5
- Intersection:
T1, T3
- Support = 2
Frequent itemset: **Milk, Bread**

Milk $\cap$ Butter

- TID(Milk) = T1, T3, T4
TID(Butter) = T2, T3, T4, T5
- Intersection:
T3, T4
- Support = 2
Frequent itemset: **Milk, Butter**

Bread $\cap$ Butter

- TID(Bread) = T1, T2, T3, T5
TID(Butter) = T2, T3, T4, T5
- Intersection:
T2, T3, T5
- Support = 3
Frequent itemset: **Bread, Butter**

Step 3: Generate Frequent 3-Itemsets

- Milk $\cap$ Bread $\cap$ Butter
- TID(Milk, Bread) = T1, T3
TID(Butter) = T2, T3, T4, T5
- Intersection:
T3
- Support = 1
- Since support $< 2 \rightarrow$ Not frequent

Mining Closed and Max Patterns

- In frequent itemset mining, the number of frequent itemsets can be very large.

To reduce redundancy, two important concepts are used:

- **Closed Frequent Itemsets**
- **Maximal Frequent Itemsets**
- These methods help to **reduce the number of patterns** while preserving useful information.

1. Closed Frequent Itemsets

- A **frequent itemset X** is called **closed** if:
- There is **no superset of X** having the **same support**.
- Closed itemsets preserve **complete support information**.

Example

If:

{Milk, Bread} support = 3

{Milk, Bread, Butter} support = 3

Then:

{Milk, Bread} is **not closed** because its superset has the same support.

2. Maximal Frequent Itemsets

- A **frequent itemset X** is called **maximal** if:
- There is **no frequent superset** of X .
- Maximal itemsets give the **largest frequent patterns** but do not preserve exact support of subsets.

Example Problem

Consider the transaction database:

Transaction ID	Items
T1	A, B, C
T2	A, B
T3	A, B, C
T4	A, B, D
T5	B, C

Minimum Support Count = **2**

Find:

- Frequent itemsets
- Closed frequent itemsets
- Maximal frequent itemsets

Step 1: Find Frequent 1-Itemsets

Count support:

Item	Support
A	4
B	5
C	3
D	1

- D is removed (support < 2)
- Frequent 1-itemsets: A, B, C

Step 2: Frequent 2-Itemsets

- Possible pairs:
- **A, B**
- Transactions: T1, T2, T3, T4
Support = 4
- **A, C**
- Transactions: T1, T3
Support = 2
- **B, C**
- Transactions: T1, T3, T5
Support = 3
- All are frequent.

Step 3: Frequent 3-Itemsets

- **A, B, C**
- Transactions: T1, T3
Support = 2
Frequent.

Step 4: Identify Closed Itemsets

Check if any superset has the same support.

- A (support 4)
Superset A,B has support 4 → Not closed
- B (support 5)
No superset with support 5 → Closed
- C (support 3)
Superset B,C has support 3 → Not closed
- A,B (support 4)
No superset with support 4 → Closed
- A,C (support 2)
Superset A,B,C has support 2 → Not closed
- B,C (support 3)
No superset with support 3 → Closed
- A,B,C (support 2)
No superset → Closed

Closed Frequent Itemsets:

B

A,B

B,C

A,B,C

Step 5: Identify Maximal Itemsets

- Maximal means **no frequent superset exists**.
- A,B,C $\rightarrow$ No superset $\rightarrow$ Maximal
- B,C $\rightarrow$ Has superset A,B,C $\rightarrow$ Not maximal
- A,B $\rightarrow$ Has superset A,B,C $\rightarrow$ Not maximal
- **Maximal Frequent Itemset:**
- A,B,C

Final Answer

- Frequent Itemsets:
A, B, C
A,B
A,C
B,C
A,B,C
- Closed Frequent Itemsets:
B
A,B
B,C
A,B,C
- Maximal Frequent Itemsets:
A,B,C

Step 1: Find Support of A

- From the table:
- Transactions containing **A**:
- T1
- T2
- T3
- T4
- So,
- $\text{Support}(A)=4$

Step 2: Find Supersets of A

- A **superset** means a larger itemset that contains A.
- Possible supersets:
 - {A, B}
 - {A, C}
 - {A, B, C}

Step 3: Check Support of Superset {A, B}

- Transactions containing **A and B**:
- T1
- T2
- T3
- T4
- So,
- $\text{Support}(A,B)=4$

Step 4: Apply Closed Itemset Rule

- Definition:
An itemset is **closed** if **no superset has the same support**.
- Here:
- $\text{Support}(A) = 4$
- $\text{Support}(A,B) = 4$
- A superset **$\{A,B\}$** has the **same support** as $\{A\}$.
- This means:
Whenever **A appears**, **B also appears** in those transactions.
- So **A alone does not give any extra information**.

- Therefore:
- A is NOT a closed itemset
- because a **superset {A,B} exists with the same support (4).**

- If:
- $\text{Support}(X) = \text{Support}(X,Y)$
Then:
- X is **not closed**

- **Step 1: Find Support of B**
- From the transactions:
- Transactions containing **B**:
- T1
- T2
- T3
- T4
- T5
- So,
- $\text{Support}(B)=5$

Step 2: Find Supersets of B

- Possible supersets of B are:
- {A, B}
- {B, C}
- {A, B, C}
- Now check their supports:
- $\text{Support}(A, B) = 4$
- $\text{Support}(B, C) = 3$
- $\text{Support}(A, B, C) = 2$

- **Step 3: Apply Closed Itemset Rule**
- Definition:
An itemset is **closed** if **no superset has the same support**.
- Here:
- $\text{Support}(B) = 5$
- All supersets have support less than 5
- **So no superset of B has the same support.**
- B is a closed itemset

- If:
- $\text{Support}(X) > \text{Support}(\text{all supersets of } X)$
- Then:
- X is **Closed**.

Pattern mining in multilevel and multidimensional space

- In real databases, data may be organized in:
- **Multiple levels of abstraction** (hierarchies)
- **Multiple dimensions or attributes**
- Pattern mining in such environments helps discover **more meaningful and flexible patterns**.

1. Multilevel Pattern Mining

- **Multilevel pattern mining** finds frequent patterns at **different levels of abstraction** using a **concept hierarchy**.
- **Example hierarchy:**
 - Food
 - Dairy
 - Milk
 - Food
 - Bakery
 - Bread

Why Multilevel Mining is Needed

- High-level patterns have **higher support**.
- Low-level patterns provide **more detailed information**.
- Different **minimum support thresholds** may be used at different levels.

Example Problem

Consider the transactions:

TID	Items Purchased
T1	Milk, Bread
T2	Milk, Cake
T3	Bread, Cake
T4	Milk, Bread
T5	Milk, Cake

Concept hierarchy:

Milk → Dairy

Bread → Bakery

Cake → Bakery

Minimum Support = 2

Find frequent patterns at:

1. Level 1 (Dairy, Bakery)

2. Level 2 (Milk, Bread, Cake)

Step 1: Convert Transactions to Level 1

TID	Items Purchased
T1	Milk, Bread
T2	Milk, Cake
T3	Bread, Cake
T4	Milk, Bread
T5	Milk, Cake

TID	Items (Level 1)
T1	Dairy, Bakery
T2	Dairy, Bakery
T3	Bakery
T4	Dairy, Bakery
T5	Dairy, Bakery

Concept hierarchy:

Milk → Dairy

Bread → Bakery

Cake → Bakery

Minimum Support = 2

Find frequent patterns at:

1.Level 1 (Dairy, Bakery)

2.Level 2 (Milk, Bread, Cake)

Step 2: Count Support at Level 1

Item	Support
Dairy	4
Bakery	5

Frequent itemsets:
Dairy, Bakery

2-Itemset

- Dairy, Bakery appears in:
T1, T2, T4, T5
- Support = 4 → Frequent

Step 3: Drill Down to Level 2

Original items:

Item	Support
Milk	4
Bread	3
Cake	3

- All are frequent.

Step 4: Find Frequent 2-Itemsets

- Milk, Bread $\rightarrow$ T1, T4 $\rightarrow$ Support = 2
Milk, Cake $\rightarrow$ T2, T5 $\rightarrow$ Support = 2
Bread, Cake $\rightarrow$ T3 $\rightarrow$ Support = 1 (Not frequent)

- Frequent 2-itemsets:

Milk,Bread

Milk,Cake

TID

T1

T2

T3

T4

T5

Items Purchased

Milk, Bread

Milk, Cake

Bread, Cake

Milk, Bread

Milk, Cake

Final Answer

Level 1 Frequent Patterns:

- Dairy
- Bakery
- Dairy, Bakery

Level 2 Frequent Patterns:

- Milk
- Bread
- Cake
- Milk, Bread
- Milk, Cake

Multidimensional Pattern Mining

- **Multidimensional pattern mining** finds patterns involving **more than one attribute (dimension)**.
- **Example dimensions:**
 - Age
 - Income
 - Product
 - Location
- **Example pattern:**
Age = Young AND Product = Milk

Types of Multidimensional Patterns

- **Inter-dimension patterns**
Each dimension appears only once.
Example: Age=Young, Product=Milk
- **Hybrid-dimension patterns**
A dimension may appear more than once.

Problem

A store records the following customer data:

Transaction ID	Age Group	Income Level	Product Purchased
T1	Young	High	Milk
T2	Young	Medium	Bread
T3	Middle	High	Milk
T4	Young	High	Milk
T5	Middle	Medium	Bread

- Minimum Support Count = **2**
- Find the **frequent multidimensional patterns**.

Step 1: Identify Dimensions

- Dimensions in the table:
- Age Group
- Income Level
- Product Purchased
- A multidimensional pattern combines values from different dimensions.
- Example:
- Age = Young AND Product = Milk

Step 2: Count Support of Single-Dimension Patterns

Age Group

- Young → T1, T2, T4 → Support = 3
Middle → T3, T5 → Support = 2
- Both frequent.

Transaction ID	Age Group	Income Level	Product Purchased
T1	Young	High	Milk
T2	Young	Medium	Bread
T3	Middle	High	Milk
T4	Young	High	Milk
T5	Middle	Medium	Bread

Income Level

- High $\rightarrow$ T1, T3, T4 $\rightarrow$ Support = 3
Medium $\rightarrow$ T2, T5 $\rightarrow$ Support = 2
- Both frequent.
- **Product**
- Milk $\rightarrow$ T1, T3, T4 $\rightarrow$ Support = 3
Bread $\rightarrow$ T2, T5 $\rightarrow$ Support = 2
- Both frequent.

Transaction ID	Age Group	Income Level	Product Purchased
T1	Young	High	Milk
T2	Young	Medium	Bread
T3	Middle	High	Milk
T4	Young	High	Milk
T5	Middle	Medium	Bread

Step 3: Generate Two-Dimensional Patterns

- **Age = Young AND Product = Milk**

- Transactions:
T1, T4

- Support = 2 → Frequent

- **Age = Young AND Income = High**

- Transactions:
T1, T4

- Support = 2 → Frequent

Transaction ID	Age Group	Income Level	Product Purchased
T1	Young	High	Milk
T2	Young	Medium	Bread
T3	Middle	High	Milk
T4	Young	High	Milk
T5	Middle	Medium	Bread

- **Income = High AND Product = Milk**

- Transactions:
T1, T3, T4

- Support = 3 → Frequent

- **Age = Middle AND Product = Bread**

- Transactions:
T5

- Support = 1 → Not frequent

Transaction ID	Age Group	Income Level	Product Purchased
T1	Young	High	Milk
T2	Young	Medium	Bread
T3	Middle	High	Milk
T4	Young	High	Milk
T5	Middle	Medium	Bread

- **Step 4: Generate Three-Dimensional Patterns**
- **Age = Young AND Income = High AND Product = Milk**
- Transactions:
T1, T4
- Support = 2 → Frequent

Transaction ID	Age Group	Income Level	Product Purchased
T1	Young	High	Milk
T2	Young	Medium	Bread
T3	Middle	High	Milk
T4	Young	High	Milk
T5	Middle	Medium	Bread

Final Frequent Multidimensional Patterns

Single dimension:

- Age = Young
- Age = Middle
- Income = High
- Income = Medium
- Product = Milk
- Product = Bread

Two dimensions:

- Age = Young, Product = Milk
- Age = Young, Income = High
- Income = High, Product = Milk

Three dimensions:

- Age = Young, Income = High, Product = Milk

1. Support

- Support measures **how frequently an itemset appears in the dataset.**
- It indicates the **popularity of the rule.**

$$\text{Support}(X \Rightarrow Y) = \frac{\text{Number of transactions containing } (X \cup Y)}{\text{Total number of transactions}}$$

2. Confidence

- Confidence measures **how often items in Y appear in transactions that contain X**.
- It indicates the **reliability of the rule**.

$$Confidence(X \Rightarrow Y) = \frac{Support(X \cup Y)}{Support(X)}$$

3. Lift

- Lift shows **how much more likely** items occur together compared to random chance. Determines the **strength and usefulness** of a rule.

$$Lift(X \Rightarrow Y) = \frac{Confidence(X \Rightarrow Y)}{Support(Y)}$$

OR

$$Lift = \frac{Support(X \cup Y)}{Support(X) \times Support(Y)}$$

- **Interpretation:**

- Lift = 1 $\rightarrow$ No relationship
- Lift > 1 $\rightarrow$ Positive relationship (strong association)
- Lift < 1 $\rightarrow$ Negative relationship

- **Example:**

If Confidence = 0.67 and Support(Bread) = 0.4, then

Lift = 0.67 / 0.4 = 1.675 (strong association)

4. Leverage

- Leverage measures **the difference between the observed frequency of X and Y appearing together and the expected frequency if they were independent.**

$$\textit{Leverage}(X \Rightarrow Y) = \textit{Support}(X \cup Y) - [\textit{Support}(X) \times \textit{Support}(Y)]$$

- Interpretation:
- Value near 0 $\rightarrow$ Independent
- Positive value $\rightarrow$ Positive relationship

5. Conviction

- Conviction measures **how strongly X implies Y**, considering how often the rule makes incorrect predictions.

$$Conviction(X \Rightarrow Y) = \frac{1 - Support(Y)}{1 - Confidence(X \Rightarrow Y)}$$

Interpretation:

- Conviction = 1 $\rightarrow$ Independent
- Conviction > 1 $\rightarrow$ Stronger rule

4. Accuracy

- Accuracy measures **how often a model predicts correctly**.
- Evaluates the **overall performance** of a classification model.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Example:

If a model makes 90 correct predictions out of 100, then

Accuracy = 90/100 = 90%

The following table represents transaction data of items purchased in a supermarket. **1 = Purchased, 0 = Not Purchased**

TID	Rice	Milk	Bread	Butter	Eggs
1	1	1	1	0	1
2	1	0	1	1	0
3	0	1	1	1	1
4	1	1	0	1	0
5	1	1	1	1	1
6	0	1	0	1	1
7	1	0	1	0	1
8	1	1	1	0	0
9	0	1	1	1	0
10	1	0	0	1	1

Find the following measures for each rule:

- **Support**
- **Confidence**
- **Lift**
- **Leverage**
- **Conviction**

Rule 1

- **Milk, Bread, Butter → Eggs**
- Calculate the **support** of the rule.
- Calculate the **confidence** of the rule.
- Compute **lift**, **leverage**, and **conviction**.
- Interpret whether the presence of Milk, Bread, and Butter increases the likelihood of purchasing Eggs.

Rule 2

- **Milk, Bread → Butter**
- Find the **support** and **confidence**.
- Calculate the **lift** and interpret whether the items are independent.
- Compute **leverage** and **conviction**.

Rule 3

- **Bread, Butter → Milk**
- Calculate **first support** and **confidence**.
- Find **lift** and **leverage**.
- Determine **conviction** and interpret the strength of the implication.

Rule 4

- **Milk, Butter → Bread**
- Compute **support** and **confidence**.
- Find **lift** and state whether the association is positive or negative.
- Calculate **leverage** and **conviction**.

Rule 5

- **Milk, Eggs → Bread**
- Calculate **support** and **confidence**.
- Compute **lift**, **leverage**, and **conviction**.
- Interpret whether this rule is useful for market basket analysis.

Solution

Useful Supports

Milk = 7

Bread = 7

Butter = 7

Eggs = 6

Support values:

- $\text{Support}(\text{Milk}) = 7/10 = 0.7$
 $\text{Support}(\text{Bread}) = 7/10 = 0.7$
 $\text{Support}(\text{Butter}) = 7/10 = 0.7$
 $\text{Support}(\text{Eggs}) = 6/10 = 0.6$

- **Rule 1**
- **Milk, Bread, Butter → Eggs**
- Transactions containing Milk, Bread, Butter = T3, T5, T9 = **3**
- Transactions containing Milk, Bread, Butter, Eggs = T3, T5 = **2**
- **Step 1: Support**
- $\text{Support} = 2/10 = 0.20$

Step 2: Confidence

- Confidence = $2/3 = 0.67$

Step 3: Lift

- Lift = $0.67/0.6 = 1.12$

Step 4: Leverage

- Leverage = $0.20 - (0.3 \times 0.6)$
- Support(Milk, Bread, Butter) = $3/10 = 0.3$
- Leverage = $0.20 - 0.18 = 0.02$

Step 5: Conviction

- Conviction = $(1 - 0.6)/(1 - 0.67)$
- Conviction = $0.4/0.33 = 1.21$

Interpretation

- Lift > 1 → Positive association
Milk, Bread, Butter slightly increase likelihood of Eggs.

Rule 2

Milk, Bread → Butter

Transactions with Milk & Bread =

T1, T3, T5, T8, T9 = 5

Transactions with Milk, Bread, Butter =

T3, T5, T9 = 3

Support

$$Support = 3/10 = 0.30$$

Confidence

$$Confidence = 3/5 = 0.60$$

Lift

$$Lift = 0.60/0.70 = 0.86$$

Lift < 1 → Negative association

Rule 3

Bread, Butter → Milk

Bread & Butter =

T2, T3, T5, T9 = 4

Bread, Butter, Milk =

T3, T5, T9 = 3

Support

$$Support = 3/10 = 0.30$$

Confidence

$$Confidence = 3/4 = 0.75$$

Lift

$$Lift = 0.75/0.7 = 1.07$$

Leverage

$$Leverage = 0.30 - (0.4 \times 0.7)$$

$$Leverage = 0.30 - 0.28 = 0.02$$

Conviction

$$Conviction = (1 - 0.7)/(1 - 0.75)$$

$$Conviction = 0.3/0.25 = 1.2$$

Interpretation

Lift slightly $> 1 \rightarrow$ Weak positive rule

Rule 4

Milk, Butter → Bread

Milk & Butter =

T3, T4, T5, T6, T9 = 5

Milk, Butter, Bread =

T3, T5, T9 = 3

Support

$$Support = 3/10 = 0.30$$

Confidence

$$Confidence = 3/5 = 0.60$$

Lift

$$Lift = 0.60/0.70 = 0.86$$

Negative association

Leverage

$$Leverage = 0.30 - (0.5 \times 0.7)$$

$$Leverage = -0.05$$

Conviction

$$Conviction = 0.3/0.4 = 0.75$$

Rule 5

Milk, Eggs $\rightarrow$ Bread

Milk & Eggs =

T1, T3, T5, T6 = 4

Milk, Eggs, Bread =

T1, T3, T5 = 3

Support

$$\text{Support} = 3/10 = 0.30$$

Confidence

$$\text{Confidence} = 3/4 = 0.75$$

Lift

$$\text{Lift} = 0.75/0.70 = 1.07$$

Leverage

$$\text{Leverage} = 0.30 - (0.4 \times 0.7)$$

$$\text{Leverage} = 0.02$$

Conviction

$$\text{Conviction} = 0.3/0.25 = 1.2$$

Interpretation

Lift > 1 $\rightarrow$ Useful rule

Customers buying Milk and Eggs are slightly more likely to buy Bread.

Final Solution

Rule	Support	Confidence	Lift	Leverage	Conviction
Rule 1	0.20	0.67	1.12	0.02	1.21
Rule 2	0.30	0.60	0.86	-0.05	0.75
Rule 3	0.30	0.75	1.07	0.02	1.20
Rule 4	0.30	0.60	0.86	-0.05	0.75
Rule 5	0.30	0.75	1.07	0.02	1.20